



HCEMM
Translational Medicine



**HUNGARIAN NATIONAL
LABORATORY**



Acceleration toolbox for Bioinformatics



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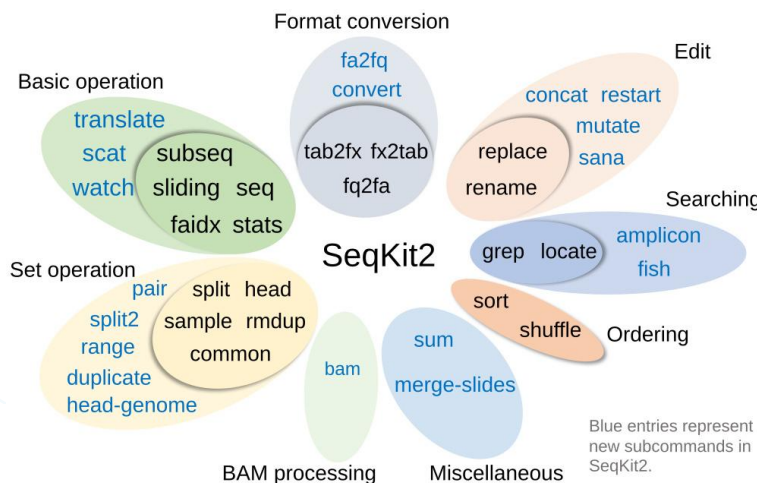
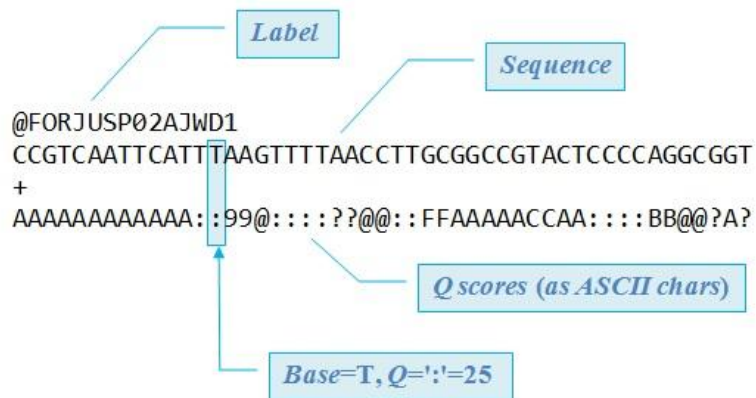
Workshop overview

- 1st day – Full Tutorial on Version control and GitHub
 - From your first repo to GH actions and pages and collaborative work
- 2nd day - Introduction to Containerization, DockerHub, and Singularity/Apptainer
 - How to build your own isolated OS for easy package management and reproducible analyses
- 3rd day – Introduction to Workflow Managers and Nextflow
 - Better reproducibility and documentation, easier management of workflow development and run
- 4th day – Introduction to Unit Testing with nf-test
 - Testing each part of pipelines so all together make a functional tool/analysis
- 5th day – Leveraging package managers and Q&A

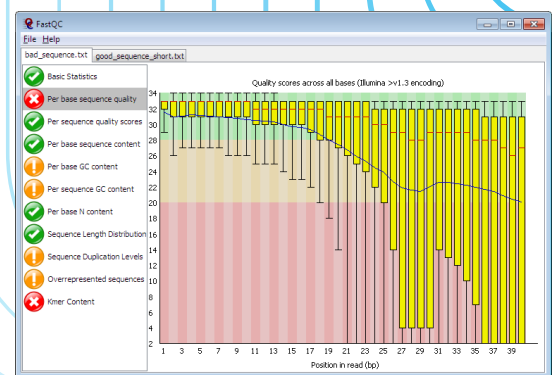
Overview of the GitHub pipeline

Your PI sent you some FastQ reads to analyze, identify presence of glucocorticoid responsive genes, and summarize all information in a web site report.

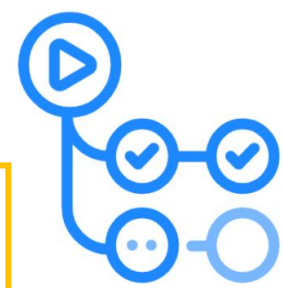
FastQ reads



Quality reports/reads summary



Pipeline testing and deployment



Annotation and reporting



GitHub Actions



Overview of the NextFlow pipeline

nextflow

Combine the execution of different tools into a formalized NextFlow application.

From FastQ reads to differential analysis

Each process only runs when needed
(parameters changed, intermediate files changed or removed)

Keep analysis functional and robust with nf-test

Biological samples/Library preparation

Sequence reads

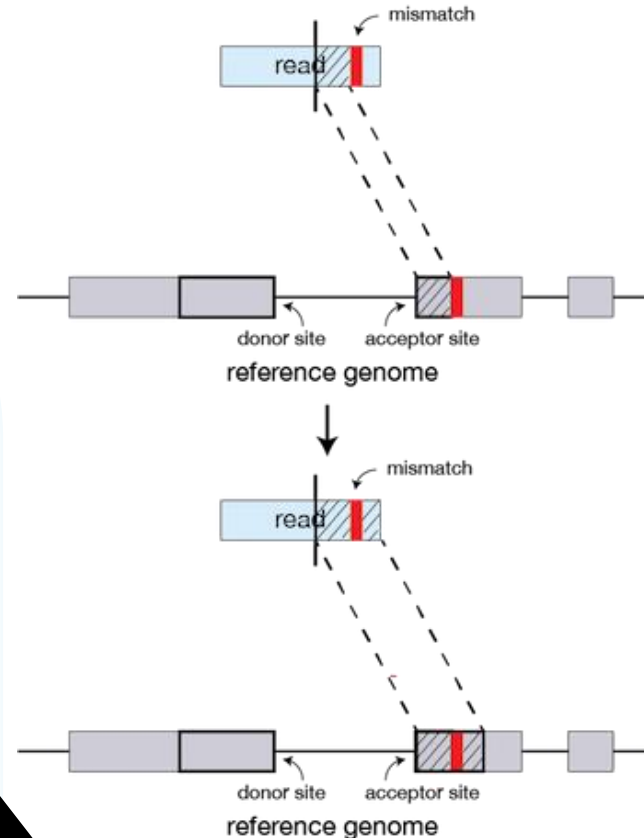
FASTQC

Adapter Trimming (Optional)

Splice-aware mapping to genome

Counting reads associated with genes

Statistical analysis to identify differentially expressed genes



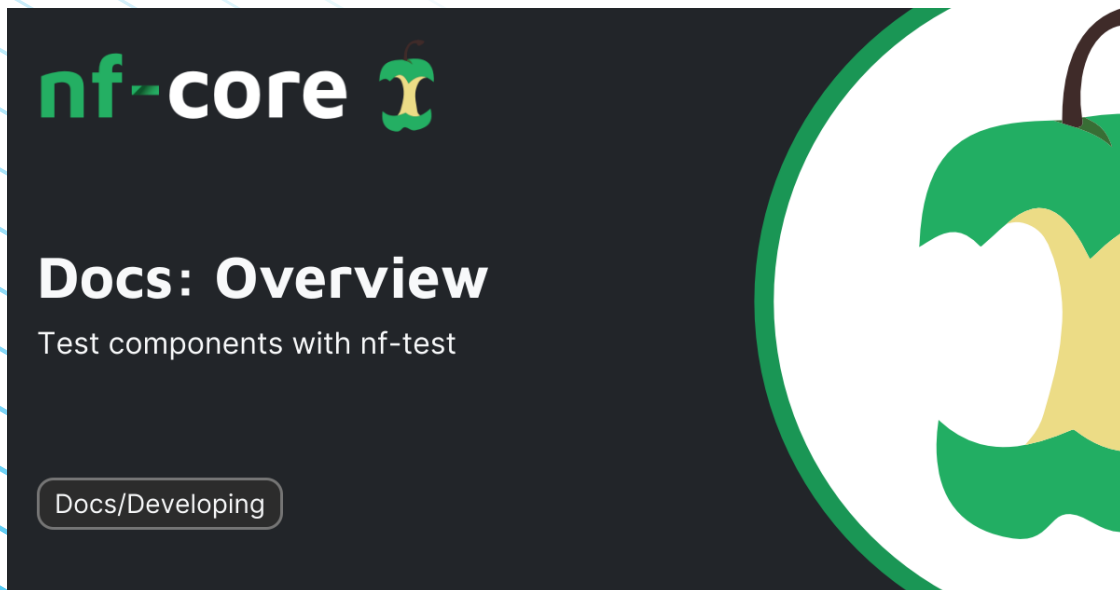
nf-test

```
nextflow_pipeline {  
  
  name "Test Hello World"  
  script "nextflow-io/hello"  
  
  test("hello world example should start 4 processes") {  
    expect {  
      with(workflow) {  
        assert success  
        assert trace.tasks().size() == 4  
        assert "Ciao world!" in stdout  
        assert "Bonjour world!" in stdout  
        assert "Hello world!" in stdout  
        assert "Hola world!" in stdout  
      }  
    }  
  }  
}
```

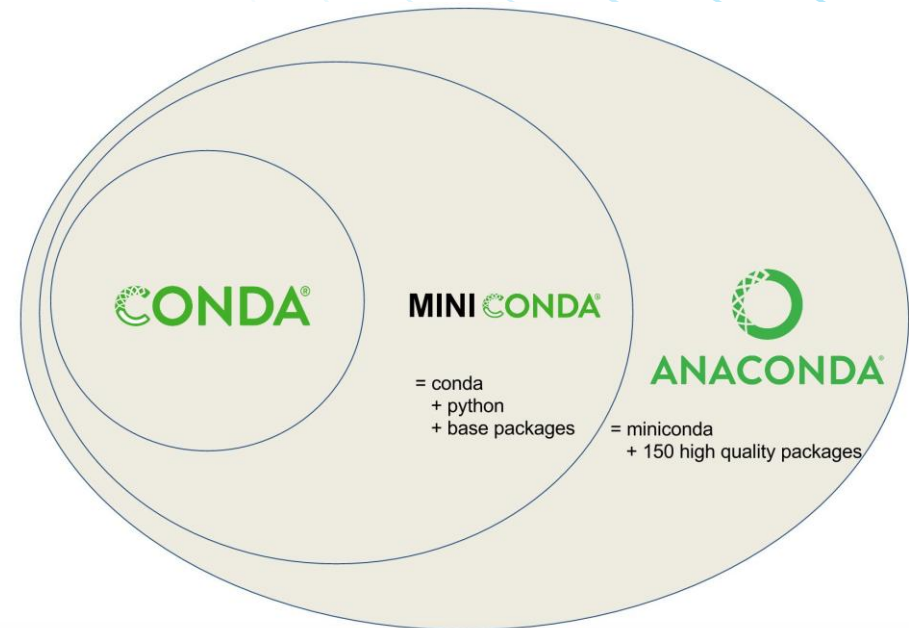
On the shoulders of little giants

Tools build on tools. Access the work of others, and publish your own.

Readily available pipelines for many case studies at nf-core



Learn about the Conda ecosystem and management of packages



Before the workshop – create a GitHub account

- GitHub (<https://github.com/>) is where modern code happens.
- Creating an account on GitHub will unlock many functionalities, some of which are required for this workshop.

Before the workshop – test access to Puli

- If you didn't access Puli before
- Attempt accessing Puli through the command: `ssh <username>@puli.hcemm.eu`
 - Change <username> to the value provided by email
 - Input the password provided by email
- If no error shows up, you have access to the supercomputer!
- Otherwise, please email us:
 - maria.kavoosi@hcemm.eu
 - joao.sequeira@hcemm.eu

Before the workshop – install VS code

- An [IDE](#) makes writing/running code easier
- This tutorial will be based on [VS code](#). You should bring a computer with it installed.
- Five groups will be composed from the 10 trainees attending the workshop. As such, it is not required to install VS code, if your group partner (chosen at the time of the workshop) has it installed.
- Still, if you intend to experiment the lessons on your system, it is highly advisable to install VS code.



See you at the workshop!

**And for any questions, don't hesitate
contacting:**

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